



Shrek On Custom VCR

A redditior crammed an entire low-res version of "Shrek" onto a single floppy disc with a 1.44MB storage size. <http://dgit.in/jan21-59>



Amazon Robotaxi

Amazon's Zoox unveils a fully-automated self-driving car for future ride-hailing services, goes upto 75 mph. <http://dgit.in/jan21-60>

important innovations in the field of robotics. During this period, Jacques de Vaucanson, a French inventor, was credited with developed three automata. The first was a robot that could play a total of twelve songs on a flute, the second one could play not only the flute but also drums and tambourines, while the third automaton was a duck that moved, quacked, flapped its wings and even ate and digested food!

In 1810, the first modern robot was designed, which was a soldier with automatic bellows (used for blowing air into fire) that were used to blow



The Automaton Trumpet Player (1810)

a trumpet instead. It was created by Friedrich Kauffman, a composer and organist from Germany. In 1898, Nikola Tesla built a remote-controlled robot torpedo, which he demonstrated at Madison Square Garden. He revealed that his wireless-controlled torpedo was based on a technology he called 'teleautomation'.

The term 'robot', however, wasn't coined until 1921 when Karel Capek, a Czech writer, used the word in a play called R.U.R. (Rossum's Universal Robots) about mechanical men that were built to work for factory assembly lines and at some point, rebelled against humans. The name is derived from the Czech word for slave. Interestingly, the word 'robotics' was also coined by a writer, Isaac Asimov, a science-fiction writer, who used the

word in his short story 'Runabout' in 1942. In his story, he viewed robots as helping servants of man and the "better, cleaner race". In 1948, Nobert Wiener laid out his principles of cybernetics. This coupled with Asimov's Three Laws of Robotics, which he laid out in 1942, laid the foundation of modern robotics.

CONTINUED DEVELOPMENT

With the arrival of programmable computers in the 1940s, robots began to develop even further. George Devol, an American inventor, invented and patented a reprogrammable robot called Unimate. In the 1960s, Joseph Engleberger, an American physicist, engineer and entrepreneur, bought out the Unimate patents from Devol and modified it into an industrial robot. He went ahead to sell it under the label 'Unimation'. He modified Devol's creation into an industrial, reprogrammable robotic arm that was ultimately installed on the assembly line of General Motors, New Jersey seven years later. The arm was used to perform repetitive and dangerous tasks on the assembly line. This was the first of countless industrial robots that are programmed for a specific function to replace unskilled labour. His work and success in robotics led him to be known as the 'the Father of Robotics'.

In the 1960s and 70s, multiple arm-like automata such as Shakey (1966), the Stanford Arm (1969), and the Silver Arm (1974) were developed for similar purposes. These, later on, gave rise to robotic technology in the medical field with automata such as Puma350 (1985) and CyberKnife (1992). Shakey, which was developed by researchers at the Stanford Research Institute, was a general-purpose robot that was able to reason concerning its own actions. Unlike other robots that would follow instructions step by step, Shaky could analyse commands and break them down into smaller chunks to perform them efficiently. The Stanford Arm which was created shortly after

became the standard for the design of future robot arms and is still being used to date.

MODERN-DAY ROBOTICS

The year 1997 was one of the most important years for robotics. It was the year of the Mars Pathfinder Mission and the robotic Mars rover, Sojourner, traversed Mars' surface, tasked with transmitting data back to Earth from the Martian soil. Then in 1999, Sony went on to release AIBO, a robotic dog with the ability to learn commands, entertain and communicate with its owner. As the technology became more robust in the 2000s, there has been more advanced automation coupled with AI and ML. For example, in 2002, SDR-4C robots were introduced that were able to remember people's faces, danced to music tracks and even sung in harmony.

Inventors also began developing humanoids that looked more and more like real humans. In recent years, artificial skin tissue, emotion, self-awareness and autonomous thinking have been introduced to the bots. One such example is Robothespian, which is a humanoid robot that can entertain and communicate with people. It is also interactive and multilingual. In 2015, Nadine was introduced, which is one of the most human-like robots ever created. It was developed by the Nanyang Technological University and it currently operates as a university receptionist. Another big achievement in robotics was the development of robotic suits named HAL (hybrid assistive limb). HAL is capable of reading brain signals and can be of great use to those with mobility issues.

Fast forward to today where robotics has replaced a large chunk of manual labour across the globe. AI is also extremely prevalent in robotics now, since robots can assess their environment, learn from their experiences, and execute tasks. The future of robotics looks bright due to the rate at which people are innovating in this burgeoning field. 

Source: AP Photo/Heinrich Sanden